Flock Energy · Data Science Take-Home Assignment

REFIT Data Cleaning and Washing-Machine Disaggregation

Name: Saumitya Srivastava
Repository: https://github.com/SaumityaSrivastava/refitwashingmachine-nilm
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Abstract

This report cleans raw REFIT smart-meter data for House 1 and analyses washing-machine use in six UK homes. It then estimates washing-machine power from the whole-home signal and evaluates the estimate on homes the model has never seen.

Part 1 (raw House 1). Raw House 1 has 8.5 million readings over 639 days. 14 % of timestamps are duplicates, 12 % of minutes fall in outages (the longest 42 days), and there are implausible and frozen readings. A documented pipeline (keep-first de-duplication, physical limits, 1-minute resampling, interpolation of gaps ≤ 5 min, outage flags) produces a clean 1-minute series.

Houses used. Part 1 uses raw **House 1** (both raw files, 2013-10-09 → 2015-07-10). Parts 2–3 use the official clean data (first 365 days per home): **Houses 2, 3, 5, 6, 7 and 10** for the EDA and as unseen test homes, and **Houses 11, 13 and 15** as extra training-only homes. **Houses 1, 4, 8, 9 and 18** were excluded because they have a second washing appliance (washer-dryer or two washing machines), whose signature would be labelled "off".

Part 2 (EDA). Across the six homes, use ranges from 1.8 to 10.4 washes a week. Washes start between 07:00 and 21:00, mostly in the morning; one home is evening-heavy and one Saturday-heavy. Median programmes range from 42 to 131 minutes.

Part 3 (model). Leave-one-house-out validation is used: test home unseen, separate validation home, three extra training homes. An ensemble of a gated LightGBM and a gated Seq2Point CNN with a cycle rule, chosen on the validation home, reaches **ON/OFF F1 0.50** (usage-profile baseline 0.16). It finds 69 % of washes, 61 % of the washes it reports are real, and its average total energy error is +9 %. For daily energy a plain LightGBM regressor is most reliable: 0.35 kWh/day error vs 0.51 for always-off.

Main error sources. Overlapping 2 kW appliances (tumble dryer, dishwasher), short cold washes, and label artefacts.

Part 4 (implications). The most realistic use is aggregated insight and targeting of evening-peak load shifting, measured with an A/B test. Applying the model to India would need local data.

Key results (unseen homes, mean over 6 test homes)

Model	F1 (ON/OFF)	Cycle precision	Cycle recall	Daily energy MAE
Always-off baseline	0.00	–	0.00	0.511 kWh
Hour-of-week profile baseline	0.16	0.35	0.79	0.484 kWh
LightGBM regressor	0.35	0.37	0.87	**0.346 kWh**
Final: ensemble (gated LightGBM + gated CNN + cycle rule)	**0.50**	**0.61**	0.69	0.433 kWh

Houses and per-home results (final model on each unseen home)

House	WM channel	Period	Cycles	Cycles / week	Median duration	F1 (ON/OFF)	Daily energy MAE
2	Appliance2	2013-09-17 → 2014-09-17	164	4.78	94.5 min	0.26	0.215 kWh
3	Appliance6	2013-09-25 → 2014-09-25	386	8.53	64 min	0.57	0.377 kWh
5	Appliance3	2013-09-26 → 2014-09-26	396	8.32	42 min	0.16	0.541 kWh
6	Appliance2	2013-11-28 → 2014-11-28	79	1.76	131 min	**0.74**	0.047 kWh
7	Appliance5	2013-11-01 → 2014-11-01	473	10.37	59 min	0.73	0.345 kWh
10	Appliance5	2013-11-20 → 2014-11-20	395	8.40	120 min	0.54	1.072 kWh
11, 13, 15	training only (never tested)						
1, 4, 8, 9, 18	excluded: second washing appliance in the home						

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1. Introduction

1.1 Problem

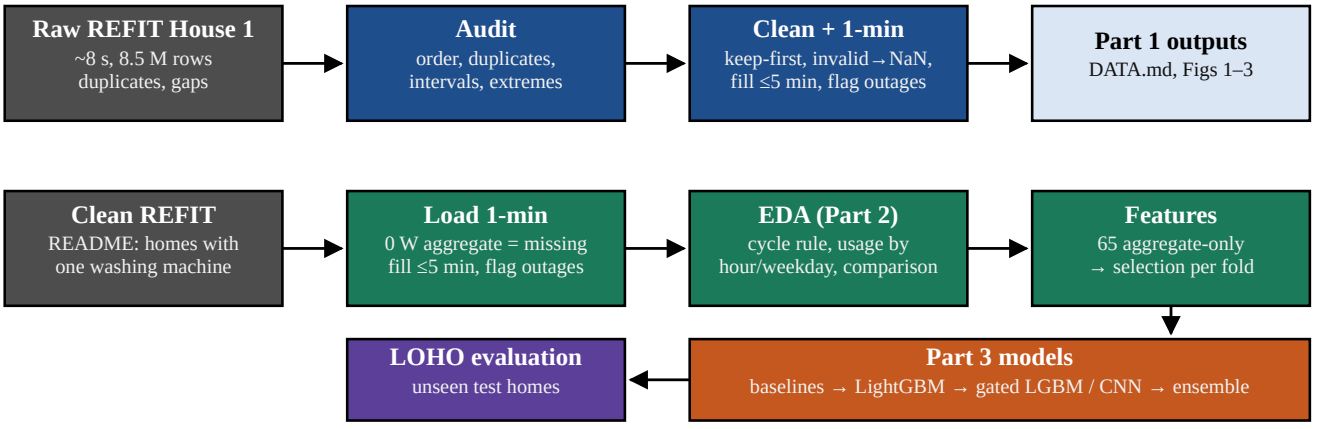
Utilities usually see only the **whole-home (aggregate) power** of a household. Knowing how much individual appliances use would let them give customers specific feedback ("your washing machine used 12 kWh last month, mostly in the evening peak") and design demand-response programmes. Estimating appliance-level consumption from the aggregate signal alone is called **non-intrusive load monitoring (NILM)** or **energy disaggregation** [2].

This report covers the whole take-home assignment:

Part	Question	Data used
1	Is raw smart-meter data clean? Build a documented cleaning pipeline	**raw** REFIT, House 1
2	How, when and how long are washing machines used in different homes?	**clean** REFIT, 6 homes
3	Can washing-machine power be estimated from the aggregate alone, for a home the model has never seen?	**clean** REFIT, 9 homes (6 tested)
4	What can a utility realistically do with this, and what are the limits?	results of Parts 2–3

The washing machine is a good but hard target. It is a large, flexible load, which makes it useful for load shifting. It runs only 2–10 % of the time, and its electrical signature changes with the programme. The ≈2 kW heater block looks the same as the ones from dishwashers, tumble dryers and kettles.

1.2 Pipeline overview



Pipeline overview (one notebook, end-to-end)

Everything runs end-to-end in one Colab notebook (`notebooks/REFIT_full_pipeline.ipynb`). The notebook:

- downloads the official archives;
- extracts only the files it needs;
- reads the appliance names from the official README;
- caches each finished cross-validation fold;
- writes all tables and figures to Google Drive.

2. Dataset

REFIT Electrical Load Measurements (University of Strathclyde) [1] covers 20 households in Loughborough, UK, recorded between October 2013 and June 2015. Each household has:

- one **aggregate** channel (whole-home active power), and
- nine plug-level **appliance** channels (`Appliance1`–`Appliance9`, individual appliance monitors), all in **watts**,
- logged roughly every **8 seconds** (median interval in raw House 1: 7 s).

The CSV files contain only `Appliance1..9`. Which appliance each channel holds is listed per house in the official README (`CLEAN_READ_ME_081116.txt`), which the notebook parses automatically.

Archive	Content	Used for
`REFIT_RAW_081116.7z` (666 MB, 40 files)	raw logger data, two parts per house	Part 1: `RAW_House1_Part1.csv` + `RAW_House1_Part2.csv`
`CLEAN_REFIT_081116.7z` (490 MB, 20 files)	official cleaned data, one file per house	Parts 2–3: Houses 2, 3, 5, 6, 7, 10, 11, 13, 15
`CLEAN_READ_ME_081116.txt`	channel → appliance mapping per house	automatic washing-machine channel selection

What the official cleaning did (dataset page): duplicate timestamps were merged; plug readings above 4 kW were set to 0 W; gaps shorter than 2 minutes were forward-filled; **gaps longer than 2 minutes were written as 0 W**. The last point matters. In the clean files a zero often means *no data*, not *appliance off*, so our loader treats 0 W aggregate minutes as missing (§4.2).

Known limitations of the data: long recording gaps (weeks in some homes); plug monitors that were unplugged or changed appliance during the study; unmetered loads (lighting, oven, hob, electric shower) that appear only in the aggregate; and some homes with two washing appliances (§4.1).

3. Part 1 — Raw House 1: inspection and cleaning

3.1 What was checked and why

Check	How	Why it matters
Timestamp parsing	`Unix` seconds → UTC; unparseable values counted	local UK time repeats one hour every October (end of summer time)
Ordering	count of negative steps between consecutive timestamps	resampling and rolling windows assume sorted data
Duplicates	duplicated timestamps, and fully duplicated rows	double counting of energy
Irregular intervals / gaps	distribution of time differences; gaps > 60 s, > 5 min, > 1 h	defines what can be filled and what must be flagged
Missing values	NaN per channel	—
Impossible / extreme values	negative power; aggregate > 20 kW; plug > 4 kW	a UK domestic supply (60–100 A × 230 V) cannot exceed ≈ 14–23 kW; plug monitors are rated ≈ 3 kW (13 A), and the official clean release uses the same 4 kW limit
Consistency	rows where the sum of the sub-meters exceeds the aggregate	sensor or timing problems

3.2 Findings

Check	Result (raw House 1, both parts)
Rows	**8,518,763** (Part 1: 4,377,877; Part 2: 4,140,886)
Period	2013-10-09 14:06 → 2015-07-10 11:56 UTC (**638.9 days**)
Unparseable timestamps	0
Out-of-order steps	2
Duplicate timestamps	**1,211,048 (14.2 %)**: 884,707 are exact duplicate rows; ≈ 326k carry *different* values
Sampling interval	median 7 s, 95th percentile 13 s
Gaps > 60 s / > 5 min / > 1 h	4,040 / 390 / 21
Longest gap	**998 h ≈ 41.6 days** (from 2014-01-24)
Missing values	aggregate 188,973 rows; appliances 232,752–510,963 per channel (Appliance 9: 1,911,811)
Negative power	none
Aggregate > 20 kW	127 readings
Plug > 4 kW	36–2,085 per channel (most on the washer-dryer, Appliance 4)
Sub-meters > aggregate	50,534 rows

House 1 channels (README): 1 Fridge, 2 Freezer (1), 3 Freezer (2), 4 Washer Dryer, **5 Washing Machine**, 6 Dishwasher, 7 Computer, 8 Television Site, 9 Electric Heater.

3.3 Cleaning pipeline (documented rules)

1. **Parse** `Unix` as UTC. Local time (`Europe/London`) is used only for hour/day features and plots.
2. **Sort** by timestamp with a stable sort.
3. **De-duplicate: keep the first reading** of each timestamp. Later copies are treated as logger re-transmissions. This is a stated convention: for the ≈ 326k conflicting duplicates there is no way to know which value is right.
4. **Impossible values → missing**: negative power, aggregate > 20 kW, plug > 4 kW.

5. **Resample to 1-minute means.** This turns the irregular ~8 s stream into a regular grid that features and models can use.
6. **Interpolate short gaps only**: runs of at most **5 consecutive missing minutes**, interpolated in time. Five minutes is too short to invent or hide a washing-machine cycle (≥ 15 min).
7. **Flag longer gaps as outages** (`outage = True`) and leave them missing. They are excluded from all statistics and models, never filled.

3.4 Result

Step	Value
Duplicates removed	1,211,048
Values set to missing (impossible)	aggregate 104; appliances 33–1,541 per channel
1-minute grid	920,031 minutes
Missing minutes before filling	115,143
Minutes filled (≤ 5-min gaps)	770–3,119 per channel (aggregate 859)
Minutes flagged as outage	**114,284 (12.4 %)** in 376 periods, total 1,905 h; longest 998 h, then 208 h (2015-05-10), 180 h (2015-06-10), 105 h (2015-04-03)

So almost all missing minutes come from *long* outages. Only ≈ 0.8 % of missing minutes were short enough to be filled.

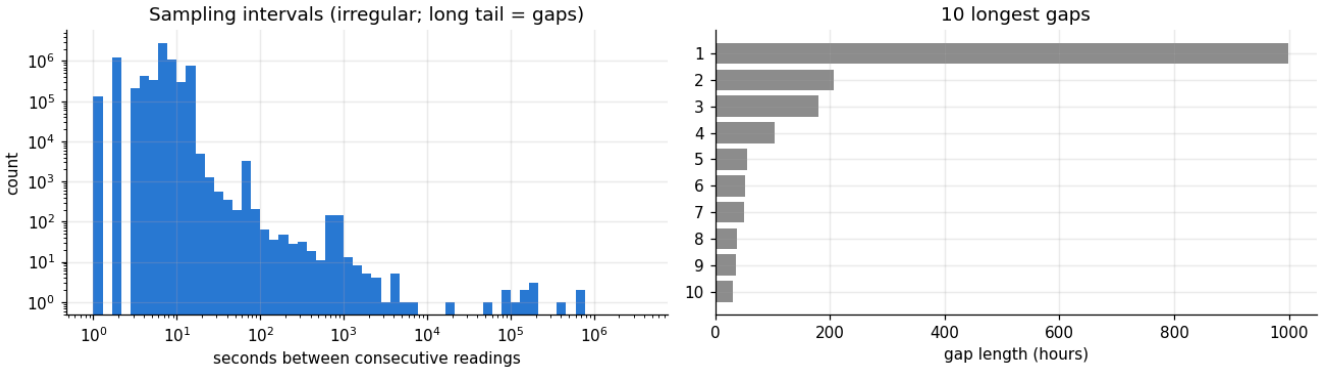


Fig. 1 – Raw House 1: distribution of sampling intervals (log scale) and the 10 longest gaps

Interpretation (Fig. 1). The interval histogram peaks at a few seconds, which is the normal logging rate. Its long tail shows real outages lasting from minutes to weeks. These must be flagged rather than interpolated, which is why the 5-minute rule exists.

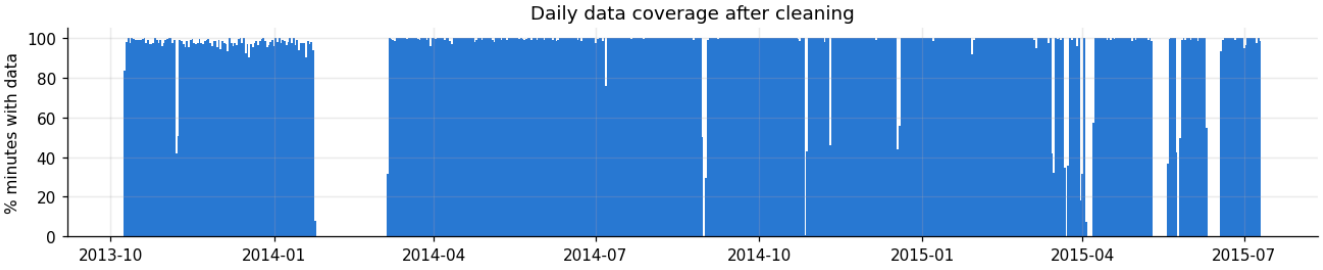


Fig. 2 – Raw House 1: share of minutes with data per day after cleaning

Interpretation (Fig. 2). Coverage is close to 100 % on most days. There is a six-week hole in early 2014 and several multi-day holes in spring 2015, so any statistic per week must be computed over *observed* time only.

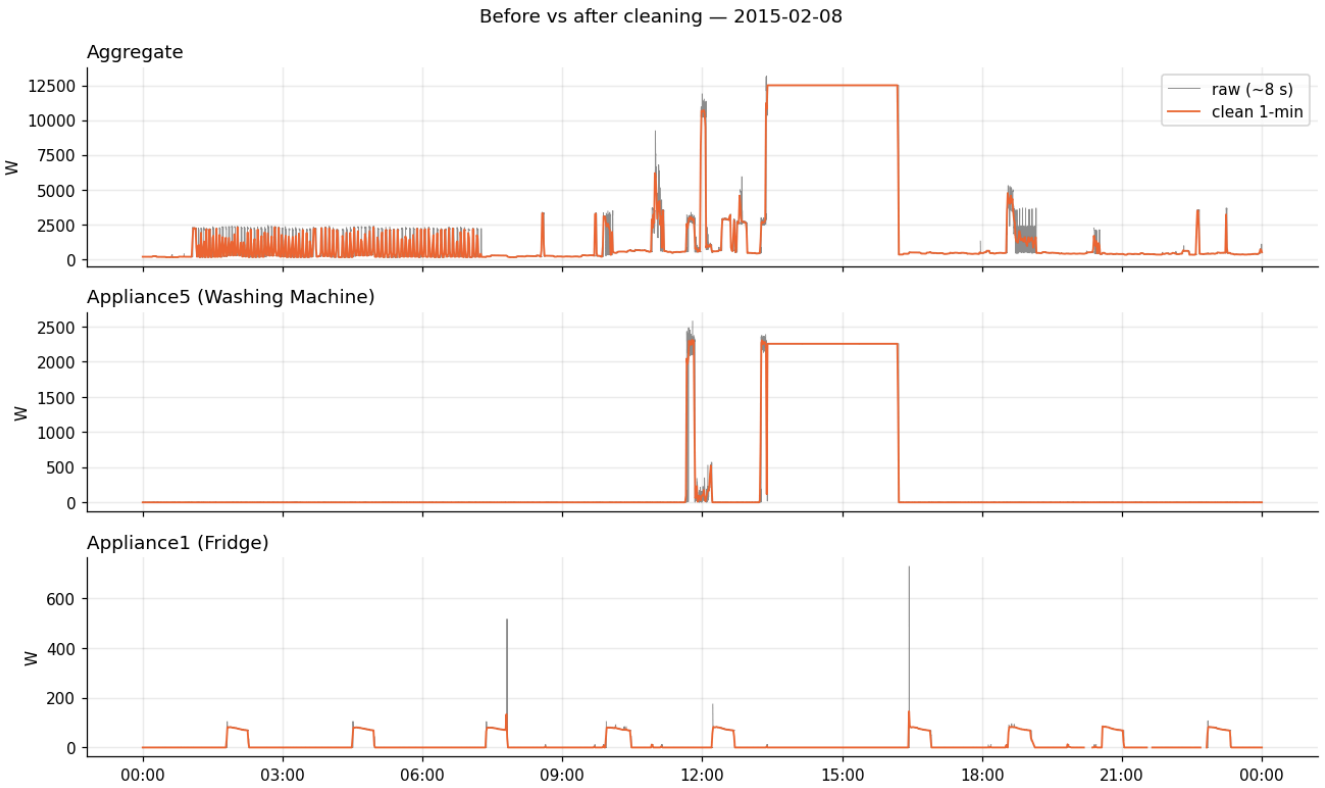


Fig. 3 – Raw House 1 before (grey, ~8 s) vs after cleaning (orange, 1 min): aggregate, washing machine, fridge

Interpretation (Fig. 3). The 1-minute means preserve every appliance event: fridge compressor cycles, the washing-machine heater block, and the evening loads. The irregular sampling disappears. The figure also reveals a problem the simple rules do not catch. On 2015-02-08 the aggregate is **frozen at ≈ 12.5 kW** and the washing machine at **≈ 2.25 kW for about 3 hours** (13:20–16:15), which is physically implausible: a logger freeze. A rule that flags runs of identical values longer than ≈ 30 minutes is an obvious next step.

3.5 Remaining concerns

- "Keep first" for conflicting duplicates is a convention, not a correction.
- Values above 4 kW were removed, so genuine short inrush peaks are lost.
- Stuck (frozen) readings are not yet detected (Fig. 3).
- House 1 has both a washer-dryer and a washing machine, which makes it unsuitable for the multi-house model (§4.1).

4. Part 2 — Washing-machine EDA (official clean data)

4.1 Which homes, and why

The README lists a washing machine in 19 homes. Homes with a **second washing appliance** were excluded automatically. In those homes the second machine produces the same signature in the aggregate but its label in the washing-machine channel is "off". For training this is label noise: the model learns to ignore real washes. For testing it creates false alarms that are not really errors.

Excluded house	Reason (README)
1, 8, 9	Washer Dryer + Washing Machine
4	Washing Machine (1) + Washing Machine (2)
18	Washer Dryer (garage) + Washing Machine

Of the eligible homes, **Houses 2, 3, 5, 6, 7, 10** are used for the EDA and as test homes, and **Houses 11, 13, 15** only as additional training homes. For each home the **first 365 days** of data are used, which keeps the runtime to about one hour on Colab and covers all seasons.

4.2 Loading the clean data

The washing-machine channel and the aggregate are read and sorted, and duplicate timestamps are dropped. **Minutes with 0 W aggregate are treated as missing**, because the official cleaning writes long gaps as 0 W and a home never draws exactly zero. The data are resampled to 1 minute. As in Part 1, gaps of at most 5 minutes are interpolated and longer ones are flagged and excluded.

4.3 Availability and data quality

Data quality

House	WM channel	Period (365 days)	Missing %	Outage periods	Longest outage (h)	WM off %
2	Appliance2	2013-09-17 → 2014-09-17	34.3	429	1488.6	95.3
3	Appliance6	2013-09-25 → 2014-09-25	13.2	91	998.4	94.4
5	Appliance3	2013-09-26 → 2014-09-26	8.7	770	40.8	93.6
6	Appliance2	2013-11-28 → 2014-11-28	14.1	97	805.3	98.0
7	Appliance5	2013-11-01 → 2014-11-01	12.5	95	998.4	92.0
10	Appliance5	2013-11-20 → 2014-11-20	9.9	777	40.8	90.7

Detected cycles

House	Cycles	Cycles/week	Median duration (min)	Median kWh/cycle	Median peak (W)	WM share of house energy %
2	164	4.78	94.5	0.574	2095	4.67
3	386	8.53	64	0.525	1934	4.16
5	396	8.32	42	0.263	2070	4.04
6	79	1.76	131	0.670	2064	1.52
7	473	10.37	59	0.519	2091	6.43
10	395	8.40	120	0.692	2080	4.32

Reading the tables.

- Missing data ranges from 9 % to 34 % (House 2 has a 62-day outage).

- The washing machine is **off 91–98 % of the time**, which shapes the choice of metrics (§5.6).
- Homes differ widely in how often they wash (1.8 to 10.4 cycles per week) and in the programmes they use.
- The washing machine accounts for 1.5–6.4 % of total household energy.

4.4 Rule for an active cycle

In plain language: the machine is **running** when its plug reads more than **20 W**. Pauses of up to **10 minutes** (soaking, drum rest) belong to the same wash. A wash counts as a cycle only if it lasts at least **15 minutes** and uses at least **0.05 kWh**.

Parameter	Value	Reason
ON threshold	20 W	above standby / sensor noise (a few W), below drum activity (≥ 50 W); similar to the low ON thresholds used for washing machines in [3]
Max pause inside a cycle	10 min	washes pause for soaking, draining and drum rest; without merging, one wash would split into many
Min duration	15 min	even quick programmes run ≥ 15 min; shorter blips are door openings or standby spikes
Min energy	0.05 kWh	removes long low-power periods (e.g. display on) that are not washes

4.5 Usage by hour of day and day of week

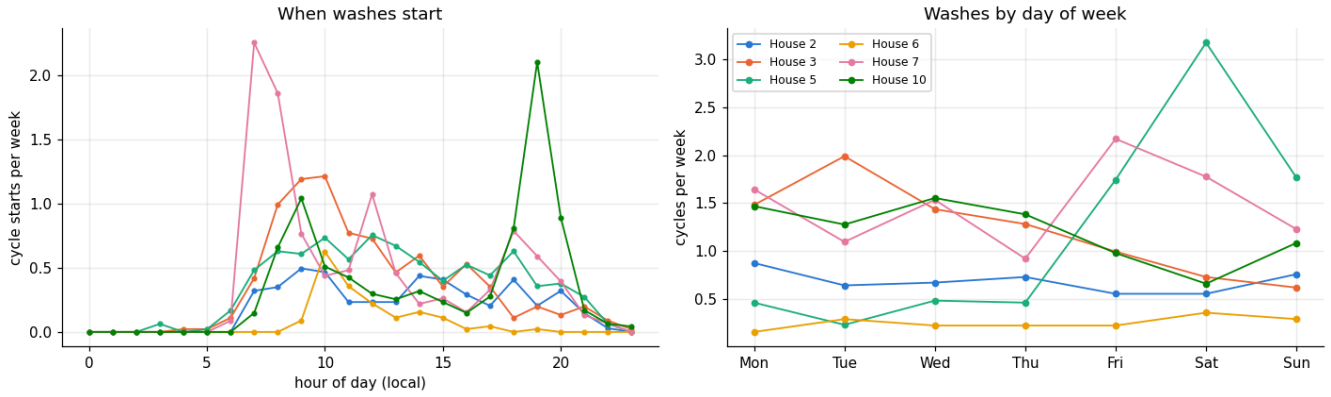


Fig. 4 – Washing-machine cycle starts per week by hour of day (left) and per weekday (right)

Interpretation (Fig. 4).

- **Time of day.** Washing is a daytime activity: almost no cycle starts between 00:00 and 05:00, and most homes start washes between 07:00 and 12:00.
  - House 7 has a sharp early peak (07–08 h, ≈ 2.2 starts/week).
  - House 3 has a late-morning peak (08–11 h).
  - House 10 is clearly **evening-heavy** (≈ 2.1 starts/week at 19:00, inside the network's evening peak).
- **Day of week.**
  - House 5 concentrates on **Saturday** (≈ 3.2 cycles/week vs ≈ 0.5 midweek).
  - House 7 rises Friday–Saturday.
  - Houses 2 and 6 wash rarely and evenly.
- **What this means:** a single "typical household" profile does not describe these homes, which is why the hour-of-week baseline is weak (§5.7).

4.6 Cross-house comparison of cycles

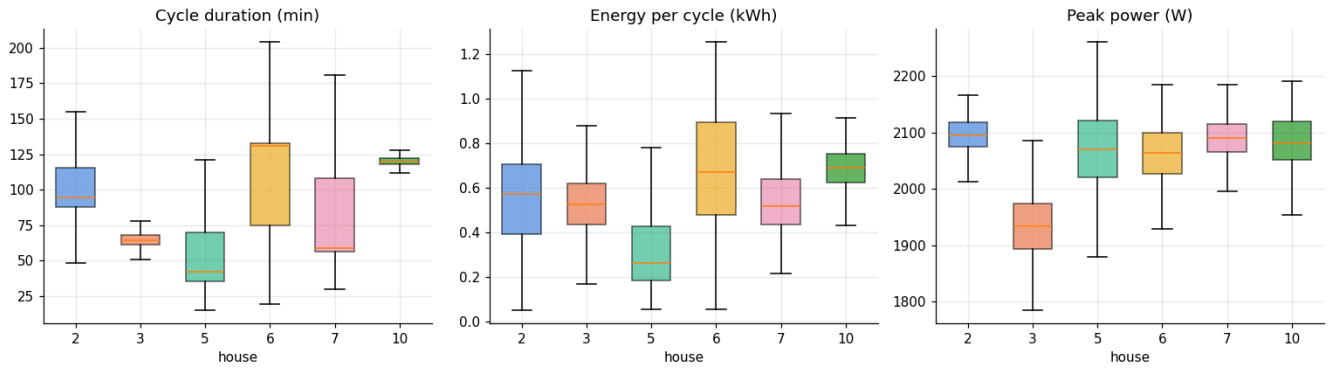


Fig. 5 – Cross-house comparison of cycle duration, energy and peak power

Interpretation (Fig. 5).

- **Peak power** is similar everywhere (≈ 2.0–2.1 kW, the heater), with House 3 a little lower (≈ 1.93 kW).
- **Programmes** differ strongly:
  - House 10 almost always runs one ≈ 120-minute programme, a very narrow box.
  - House 6 runs long cycles (median 131 min, 0.67 kWh).
  - House 5 mostly runs short, low-energy washes (median 42 min, 0.26 kWh), i.e. cold or quick programmes.
- **What this means:** a model trained on some homes sees different programme mixes in a new home, and short or cold washes are hardest to detect (§5.9).

4.7 Representative traces

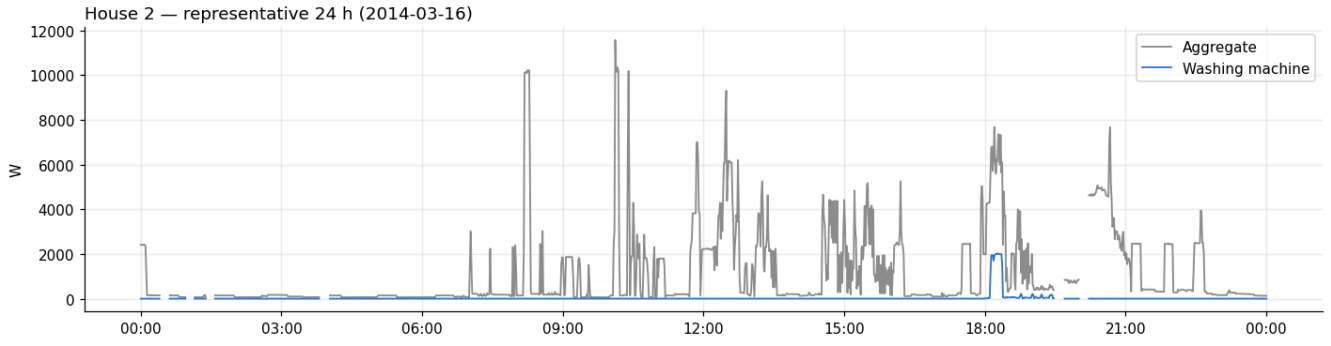


Fig. 6 – Representative 24-hour aggregate and washing-machine trace

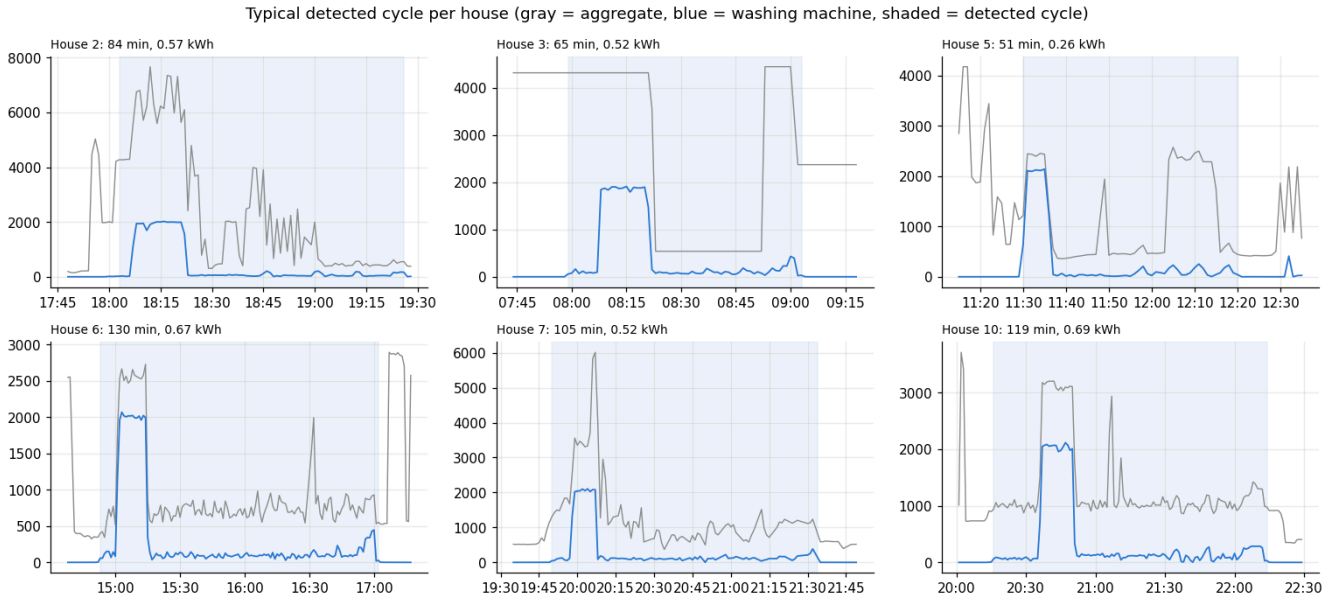


Fig. 7 – A typical detected cycle per house (shaded = detected cycle)

Interpretation (Figs. 6–7).

- **Typical cycle:** a **≈ 2 kW heater block** lasting 10–25 minutes, then 100–500 W drum and rinse activity with short pauses, then a spin.
- **What stands out in the aggregate:** the heater block is clearly visible.
- **What is hidden:** the drum phase is small compared with the fridge, lighting and cooking, and other 2 kW appliances frequently overlap.
- **Shaded windows:** the cycle rule captures the whole wash, including its pauses, as one event.

5. Part 3 — Disaggregation model

5.1 Task

For every minute t of a home, estimate the washing-machine power $\hat{y}_t$ (W) using **only the aggregate power** around t and the calendar. Sub-meter channels are never used as inputs. They are the targets, and are used for diagnosis only in §5.9. In a real deployment only the aggregate exists.

5.2 Features (65, aggregate only)

Group	Features	Purpose
Raw and steps	aggregate; 1-minute difference backward and forward; absolute difference	switching edges, as in Hart's event-based NILM [2]
Base-load removal	24-hour rolling minimum; aggregate minus base; relative excess over the 24-hour median	removes always-on loads (fridge, standby) so the model sees the *extra* load
Multi-scale centred windows (5, 15, 31, 61, 121 min)	mean, std, max, min, range	short windows see the heater block, long windows see the whole wash; centred windows follow the Seq2Point idea [4]
Window normalisation	(aggregate − window mean) / (window std + 10); aggregate − window min	makes features comparable across homes with different base loads (per-window normalisation idea of NILMFormer [6])
Event counts per window	number of jumps > 200 W; minutes with > 1.5 kW extra load; minutes with 80–800 W extra load	spin and drum activity, heater on-time
Context before / after	minutes with > 1.5 kW extra load in the 30 / 60 min before and after	the heater block comes *before* the drum phase
Calendar	hour (sine and cosine), day of week, weekend	usage habits

Centred windows use up to 60 minutes of "future" aggregate. That is fine for disaggregating historical data such as monthly reports. A real-time version would use trailing windows only.

5.3 Feature selection and reduction (per fold, training homes only)

1. **Variance filter**: drop constant features.
2. **Correlation filter**: rank features by absolute correlation with the target, then drop any feature correlated above 0.97 with one already kept. This removes redundant window statistics.
3. **Permutation importance** on the validation home, using a quick LightGBM: keep features whose shuffling increases MAE by at least 1 % of the most important feature's effect.

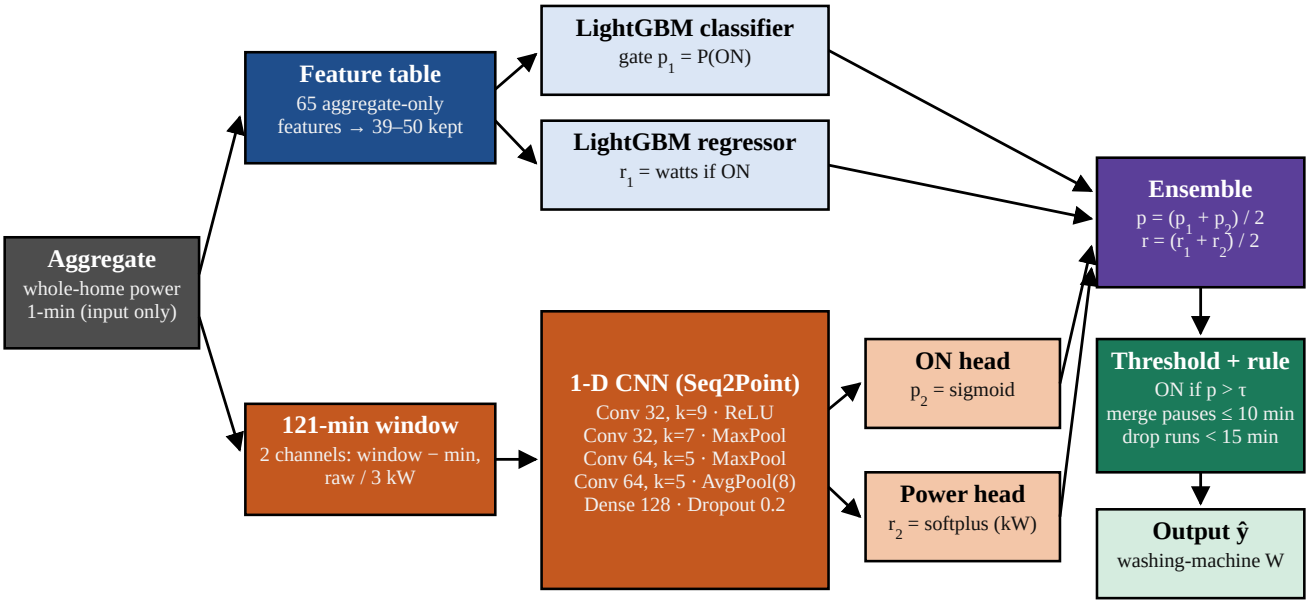
In the six folds **39–50 of the 65 features** were kept (see §5.8).

5.4 Training-set thinning

Two years of minutes × several homes is slow to train on a free Colab machine. Every "interesting" minute is kept: within ±60 min of a wash, or with more than 300 W extra load, which covers possible confounders. Of the quiet remainder, 15 % is kept at random with weight 1/0.15, so the model still sees the true OFF frequency. This cuts training time several-fold without discarding information near washes.

5.5 Models compared

Code	Model	Idea
B0	Always-off	predicts 0 W. The machine is off > 90 % of the time, so this already has a low MAE
B1	Hour-of-week profile	average washing-machine power for each weekday × hour, learned from the training homes
A	LightGBM regressor	one gradient-boosted tree model predicting watts directly
B	**Gated LightGBM**	classifier p_t = P(ON) and a regressor r_t = watts if ON (trained on ON minutes only); soft output $\hat{y}_t = p_t \cdot r_t$. Splitting "is it running?" from "how many watts?" follows subtask gated networks [5]
B+	Gated LightGBM + cycle rule	hard ON mask $p_t > \tau$ (τ tuned on the validation home), then the EDA cycle rule on the *predicted* mask: pauses ≤ 10 min merged, runs < 15 min deleted; inside merged pauses the soft value $p \cdot r$ is used
C+	Gated Seq2Point CNN + cycle rule	a 1-D CNN reads a 121-minute window of aggregate and predicts the centre minute with two heads (ON probability, watts) [4, 5]; same post-processing
E	Ensemble + cycle rule	average of B and C probabilities and watts, then post-processing
S	**Selected on the validation home (final model)**	in each fold, the candidate (A, B, B+, C+, E) with the best F1 on the *validation* home. The test home plays no part in the choice

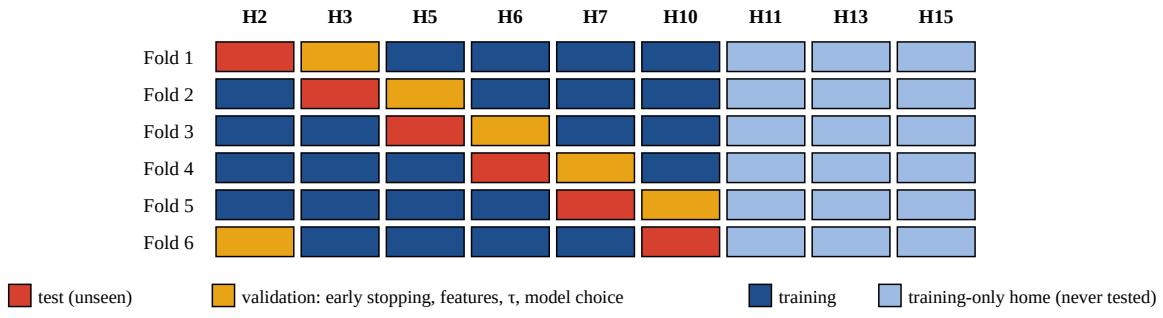


Model architecture of the final ensemble: gated LightGBM branch (top) and gated Seq2Point CNN branch (bottom), averaged and post-processed with the cycle rule

CNN architecture. The network follows Seq2Point (Zhang et al. [4]): a window of aggregate goes through convolution layers and a dense layer, and the model predicts the washing-machine power at the window's centre minute. The two output heads, ON probability and watts, follow the subtask gating of Shin et al. [5]. The layer sizes are a smaller version chosen for this data, not copied from either paper.

- **Input:** 2 channels (window minus its minimum, in kW; raw aggregate / 3 kW).
- **Layers:** Conv(32, k9) → Conv(32, k7) → MaxPool → Conv(64, k5) → MaxPool → Conv(64, k5) → AdaptivePool(8) → Dense(128) → Dropout 0.2, then two heads.
- **Loss:** binary cross-entropy on ON/OFF plus squared error on kW, counted on ON minutes only.
- **Training:** Adam (lr 10^{-3}, cosine schedule), up to 12 epochs with early stopping on validation MAE. Each epoch samples all ON minutes plus 3× as many random OFF minutes. Trained on a T4 GPU.

5.6 Validation design and leakage



Leave-one-house-out validation: every test home is unseen; validation home chooses settings and model

Leave-one-house-out (LOHO). Each of the 6 homes is the **test** home once. The next home in the list is the **validation** home, used for early stopping, feature selection, the threshold τ and the model choice S. The remaining 4 homes plus Houses 11, 13 and 15 are the **training** homes.

How this avoids leakage:

- No minute of the test home is used for training, feature selection, threshold tuning or model choice.
- Features are computed **per home**, so rolling windows never mix data from two homes.
- A random minute-level split would place neighbouring, almost identical minutes, often from the *same wash*, in both training and test, and would hugely overstate accuracy.
- LOHO measures what a utility needs: accuracy on a **new customer**.

Metrics (y = true, $\hat{y}$ = predicted watts per minute; ON = power > 20 W):

Metric	Definition	Why
MAE (W)	mean of $\lvert y - \hat{y} \rvert$	overall watt error, easy to explain; but it rewards "always off" because the machine is usually off
RMSE (W)	$\sqrt{}$ mean of $(y - \hat{y})^2$	penalises large misses
MAE when ON (W)	MAE over minutes with y > 20 W	accuracy *during* washes
Precision / Recall / F1	on the ON/OFF state per minute; F1 = 2PR / (P + R)	does the model know *when* the machine runs? Robust to the 90–98 % OFF imbalance
Energy error %	$(\Sigma\hat{y} - \Sigma y) / \Sigma y$	total over- or under-estimate, as on a bill
Daily-energy MAE (kWh/day)	mean of |daily kWh predicted − actual|	bill-level accuracy per day
Cycle recall / precision	share of true cycles overlapped by a predicted cycle / share of predicted cycles overlapping a true one (same cycle rule on both)	customer-facing "you ran 3 washes this week"

These choices follow common NILM evaluation practice [3, 7].

5.7 Results — unseen homes (mean ± std over 6 test homes)

Model	MAE (W)	MAE when ON (W)	Precision	Recall	F1	Energy err %	Daily MAE (kWh)	Cycle recall	Cycle precision
B0 Always-off	24.8 ± 10.7	404 ± 51	0.00	0.00	0.00	−100	0.511 ± 0.246	0.00	–
B1 Hour-of-week profile	41.4 ± 8.4	374 ± 51	0.09	0.73	0.16 ± 0.06	+16 ± 96	0.484 ± 0.130	0.79	0.35
A LightGBM regressor	35.2 ± 11.0	297 ± 81	0.24	0.69	0.35 ± 0.11	+19 ± 51	**0.346 ± 0.117**	**0.87**	0.37
B Gated LightGBM (soft)	37.9 ± 17.2	327 ± 80	0.23	0.65	0.33 ± 0.04	+2 ± 44	0.434 ± 0.271	0.80	0.39
B+ Gated LightGBM + cycle rule	48.8 ± 34.1	295 ± 104	0.37	0.56	0.34 ± 0.21	+67 ± 124	0.765 ± 0.533	0.66	0.39
C+ Gated CNN + cycle rule	34.8 ± 26.8	258 ± 91	0.50	0.45	0.42 ± 0.18	+24 ± 87	0.484 ± 0.437	0.72	**0.61**
E Ensemble + cycle rule	**30.2 ± 21.2**	**255 ± 103**	**0.57**	0.61	**0.50 ± 0.24**	+9 ± 79	0.433 ± 0.354	0.69	**0.61**
S Selected on validation (final)	30.2 ± 21.2	255 ± 103	0.57	0.61	**0.50 ± 0.24**	+9 ± 79	0.433 ± 0.354	0.69	0.61

Per test home.

Test home	F1: B1 profile	F1: A	F1: B+	F1: C+	F1: final (S)	Daily MAE always-off	Daily MAE A	Daily MAE final
2	0.13	0.29	0.09	0.43	**0.26**	0.269	0.230	0.215
3	0.17	0.33	0.41	0.43	**0.57**	0.571	0.389	0.377
5	0.16	0.26	0.09	0.09	**0.16**	0.605	0.501	0.541
6	0.07	0.29	0.57	0.59	**0.74**	0.148	0.214	0.047
7	0.26	0.55	0.50	0.59	**0.73**	0.729	0.300	0.345
10	0.19	0.36	0.41	0.39	**0.54**	0.741	0.443	1.072

5.8 Fold details

Test	Validation	Training homes	Features kept (of 65)	Chosen on validation	Test F1	Time (s)
2	3	5, 6, 7, 10, 11, 13, 15	45	E	0.259	475
3	5	2, 6, 7, 10, 11, 13, 15	39	E	0.568	325
5	6	2, 3, 7, 10, 11, 13, 15	42	E	0.162	480
6	7	2, 3, 5, 10, 11, 13, 15	44	E	0.741	686
7	10	2, 3, 5, 6, 11, 13, 15	50	E	0.732	408
10	2	3, 5, 6, 7, 11, 13, 15	49	E	0.543	491

The validation home picked the ensemble in **every** fold, so the final model S equals E. The choice is stable, and it was made without looking at test scores.

5.9 Interpretation

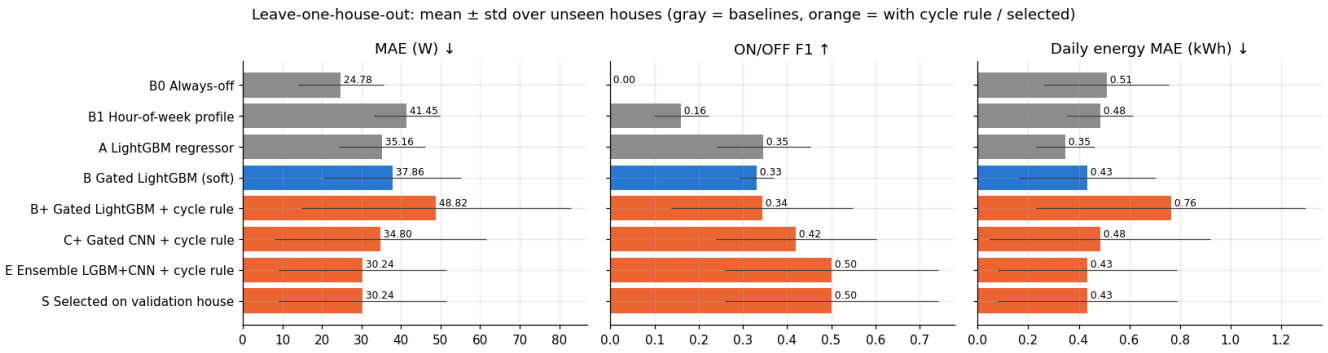


Fig. 8 – Leave-one-house-out ablation: mean ± std over the six unseen homes

- **Why "always off" has the lowest MAE but is useless.** The machine is off 91–98 % of the time, so predicting 0 W gives a small average error while finding no washes at all (F1 = 0, cycle recall = 0). The final model trades a slightly higher MAE (30 W vs 25 W) for actually finding washes: **F1 0.50 vs 0.16** for the usage-profile baseline, 69 % of real cycles found, and **61 % of predicted cycles real**. Its error *while the machine is running* is the lowest of all models (255 W vs 404 W for always-off).
- **What the ablation shows.**
  - Gating alone (B) does not help LightGBM much (F1 0.33 vs 0.35).
  - Adding the cycle rule to LightGBM (B+) raises precision (0.37) but sometimes merges false alarms into long false cycles: it has the worst energy error (+67 %) and a large spread.
  - The CNN (C+) is more precise (0.50).
  - **Averaging the two (E)** gives the best F1, precision and MAE-when-ON: the two models make different mistakes.
- **Energy.**
  - The final model's total energy error is small on average (+9 %) but varies a lot by home (± 79 %).
  - For **daily energy the plain LightGBM regressor (A) is most reliable**: 0.35 kWh/day vs 0.51 for always-off, a 32 % improvement.
  - So for a monthly bill-style estimate A is the safer choice, and for detecting *when* washes happen the ensemble is.
- **Variation between homes is large** (F1 0.16 to 0.74):
  - Homes with long, regular hot programmes are easy: House 6 (131 min), House 7, and House 10 with its single 120-min programme.
  - House 5 is hard: mostly short cold washes (median 42 min, 0.26 kWh), so the heater block is often missing.
- **Effect of excluding second-machine homes.** An earlier run on Houses 1–6 kept House 1 (washer-dryer) and House 4 (two machines) and used no extra training homes. The ensemble reached only F1 0.28 ± 0.20 there, and those two homes were the worst. The test homes differ between the runs, so this is indicative rather than a controlled comparison. It does support the exclusion rule.

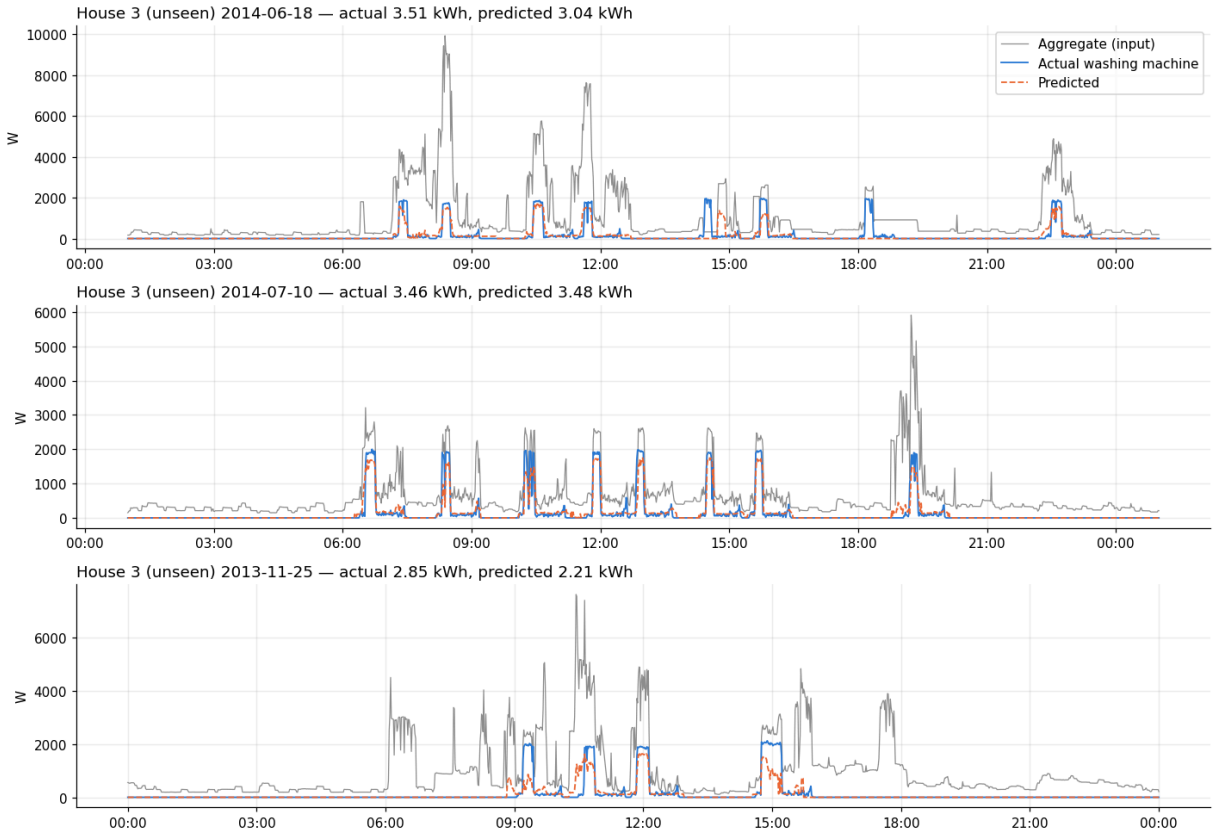


Fig. 9 – Actual vs predicted washing-machine power on unseen House 3 (median home by F1)

Interpretation (Fig. 9).

- **Which home:** House 3 is shown because it is the *median* home by F1, not the best one; the model never saw it during training.
- **What is found:** on busy days with 7–9 washes, the heater blocks and most drum phases are detected. Predicted vs actual daily energy is 3.04 vs 3.51 kWh, 3.48 vs 3.46 kWh, and 2.21 vs 2.85 kWh.
- **Where it misses:** when a wash overlaps another large load, e.g. at 15:00 on 2013-11-25 under a ≈ 4 kW block.

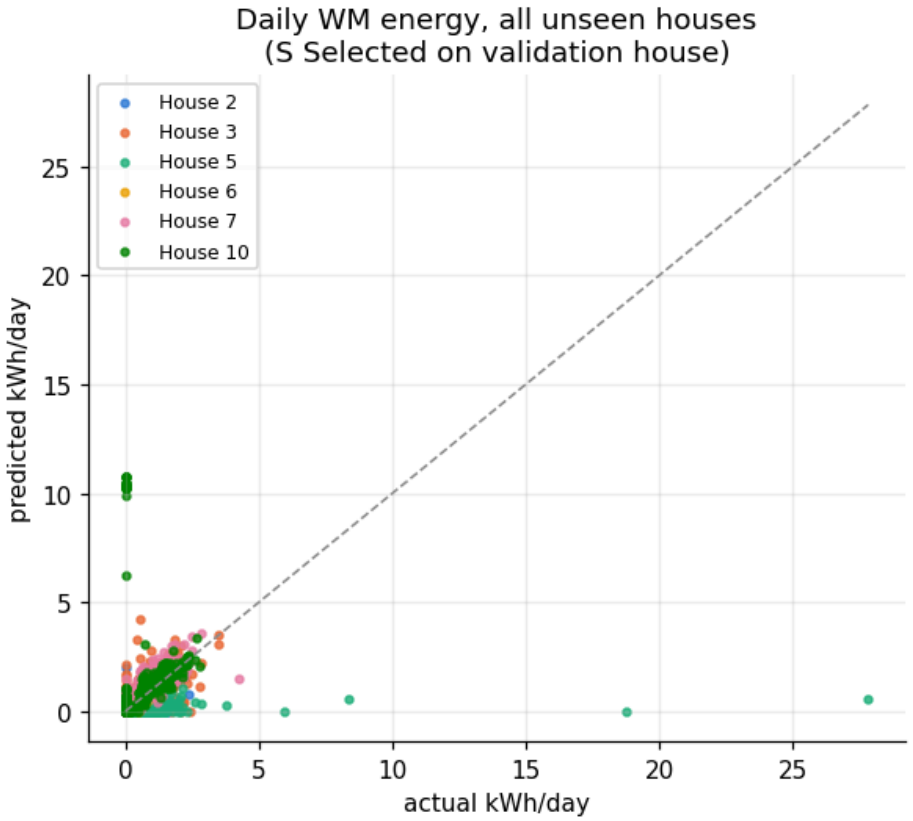


Fig. 10 – Daily washing-machine energy, predicted vs actual, all unseen homes

Interpretation (Fig. 10).

- **Normal days:** most days lie close to the diagonal up to ≈ 3 kWh/day.
- **Outliers:** the large errors come from a few days that look like **data artefacts**:
  - House 5 has days with 6–28 kWh of "washing-machine" energy. That is physically impossible for a washing machine and points to stuck plug readings.
  - House 10 has days with ≈ 10 kWh predicted but 0 kWh recorded, which looks like a plug dropout or a long confounding load.
- **Effect:** these few days dominate the daily-energy error of Houses 5 and 10.

5.10 Failure analysis

For each appliance in the test homes we compared how often it was ON during **false-alarm minutes** (model says ON, washing machine off) with how often it is ON during normal OFF time. A ratio (lift) above 1 means it is over-represented in false alarms. Sub-meters are used here only for diagnosis.

Appliance	ON during false alarms	ON during normal OFF time	Lift
Tumble Dryer	26.2 %	3.7 %	**7.0**
Dishwasher	32.6 %	6.8 %	**4.9**
Magimix (blender)	79.4 %	25.5 %	3.1
Kettle	3.0 %	1.1 %	2.8
Microwave	3.2 %	1.6 %	2.0

1. **Overlapping heating loads.** Tumble dryers, often started right after a wash, and dishwashers draw the same ≈ 2 kW heating blocks. They are the main cause of false alarms.
2. **Short or cold washes.** Of 1,893 true test cycles, **1,333 (70 %) were detected**. Missed cycles are shorter (median 59 vs 80 min) and use less energy (0.35 vs 0.59 kWh) at a similar peak. When the heater block is short or hidden under

other loads, the low-power drum phase alone is not recognised.

3. **Sensor artefacts / label noise.** Stuck plug readings (House 5) and plug dropouts (House 10) create large daily errors (Fig. 10) that are partly errors of the *label*, not the model.
4. **New homes behave differently.** With F1 from 0.16 to 0.74, a utility cannot know in advance how well the model will do for a particular customer. A few labelled weeks per home (or per machine type) would help.

5.11 Single-house check (context)

An earlier single-house experiment on raw House 1 used a chronological 70/15/15 split with a 2-day purge between splits. There the gated LightGBM with cycle rule reached F1 0.80 and daily-energy MAE 0.075 kWh (vs 0.186 kWh for always-off). The same model is much better on *later data of the same home* than on a *new home* (F1 0.50). Generalising across households is the real difficulty.

6. Part 4 — Practical implications

6.1 Opportunity: shift washes out of the evening peak

Insight. In the analysed homes almost all washes start between 07:00 and 21:00. Most homes wash in the morning, but some have a clear evening habit: House 10 starts ≈ 2 washes per week around 19:00, in the evening network peak, and House 5 concentrates its washes on Saturday. A washing machine is one of the most flexible loads in a home. Most washes can run at midday (solar), late in the evening, or overnight using the delay timer that most machines have, without changing the customer's routine.

Action. Use the disaggregated history to find customers with a recurring evening wash pattern and send them a personalised nudge, for example: "Your washes usually start around 19:00. Setting the delay timer to start after 21:00 would move about X kWh per month off-peak." Optionally add a time-of-use incentive.

How to measure whether it worked: a randomised A/B test.

1. **Eligibility:** customers with at least one detected evening wash per week in a 4-week baseline.
2. **Groups:** randomise eligible customers into **treatment** (nudge) and **control** (no nudge).
3. **Primary metric:** share of detected washing-machine energy in the 16:00–20:00 window, after 8 weeks, compared between groups as a difference-in-differences against the baseline.
4. **Secondary metrics:**
   - total washing-machine kWh, which should stay the same, because this is shifting, not saving;
   - total evening household load;
   - opt-out rate.
5. **Why the model's errors do not bias the result:** the same model scores both groups, so its systematic errors largely cancel in the comparison.
6. **Ground-truth check:** fit plug monitors in a small subsample to confirm the estimated shift.

6.2 Suitable uses and limits of the model

Suitable:

- Monthly washing-machine energy per customer, using the LightGBM regressor (≈ 0.35 kWh/day error on new homes; reliable at the month level).
- Aggregated "how often and when" insights (weekly patterns, evening vs morning).
- Targeting load-shifting campaigns.

Not suitable:

- Billing, or any decision about an individual wash.
- Homes with a second washing appliance.
- New regions or meter types without re-validation.

6.3 Limitations

- **Overlapping appliance signatures.** Dishwashers, tumble dryers, kettles and ovens produce the same ≈ 2 kW blocks. They are the main source of false alarms.
- **Missing data.** REFIT has long outages (House 1: 41 days), and the clean release writes gaps of more than 2 minutes as 0 W. Without treating those as missing, a model would learn "machine off" from missing data. Plug dropouts and stuck readings add label noise.

- **Household differences.** Machine model, programmes (cold vs hot), number of occupants and routines vary. That is why accuracy on new homes is lower and more variable than on a known home.
- **UK → India.**
  - **Different machines:** many Indian homes use semi-automatic or top-load machines **without an internal heater**, i.e. without the 2 kW block the model relies on most, and wash in cold water.
  - **Different grid:** voltage fluctuations, power cuts and inverter backups distort the aggregate.
  - **Different confounders:** large loads such as air-conditioners, geysers, water pumps and ceiling fans, with strong seasonality.
  - **Different habits:** usage times differ, e.g. linked to water-supply timings.
  - **What would be needed:** the UK model should not be applied directly. It would need retraining and re-validation on Indian labelled data (even a few dozen plug-monitored homes), probably with a lower ON threshold and different features.

6.4 What gets harder with 15- or 30-minute smart-meter data

At 1-minute resolution a wash is a 30–120-minute *shape*: a heater block, drum spikes, a spin. At 15–30-minute resolution it collapses into 2–8 averaged points. The sharp switch-on edges and the flat top of the heater block disappear, so the washing machine becomes indistinguishable from a dishwasher, tumble dryer, oven or kettle used in the same interval. Short cold or quick washes may vanish into the base load entirely. Cycle counts and start times become uncertain by at least one interval. The useful output shrinks to coarse **daily or monthly energy estimates** with wider error bars, relying more on behaviour (time of day, weekday) than on the electrical signature. Models would have to be retrained and evaluated on down-sampled data; the 1-minute results in this report would overstate their accuracy.

7. Conclusions and next steps

Conclusions.

1. Raw REFIT data need real cleaning: 14 % duplicate timestamps, 12 % outage time, implausible values and frozen readings. A simple, documented pipeline handles most of this. Long gaps are flagged, never filled.
2. Washing-machine use differs strongly between homes, in frequency (1.8–10.4 cycles per week), timing (morning vs evening vs weekend) and programme (42–131 min median duration).
3. On homes the model has never seen, an ensemble of a gated LightGBM and a gated CNN with a cycle rule detects washes far better than the baselines (F1 0.50 vs 0.16; 61 % of predicted cycles are real). A plain LightGBM regressor gives the most reliable daily energy (−32 % error vs always-off).
4. Most errors come from overlapping 2 kW appliances, short cold washes, and label artefacts.

Next steps, in order of expected value.

1. Detect stuck readings and plug dropouts, and clean the labels.
2. Use all eligible homes and the full two-year period.
3. Add hard-negative training on dishwasher and tumble-dryer periods.
4. Calibrate per home with a few labelled weeks.
5. Produce probabilistic energy estimates with uncertainty bands for customer reports.
6. Re-evaluate at 15/30-minute resolution.

8. Reproducibility and AI tools

How to run.

1. Open `notebooks/REFIT_full_pipeline.ipynb` in Google Colab.
2. Choose *Runtime → T4 GPU*.
3. Choose *Run all*.

The notebook downloads the archives and README, selects the homes automatically, and writes all outputs to `MyDrive/refit_outputs/`. Finished folds are cached, so a disconnected session resumes where it stopped. Runtime with the defaults (6 test homes + 3 training homes, 365 days each) was about one hour on a T4 GPU.

Main settings (single config cell):

Setting	**Value**
Test/EDA homes / extra training homes	6 / 3 (automatic, eligible homes only)
Days per home	first 365
Cleaning	aggregate ≤ 20 kW, plug ≤ 4 kW, fill gaps ≤ 5 min
Cycle rule	20 W, pause ≤ 10 min, ≥ 15 min, ≥ 0.05 kWh
LightGBM	3,000 trees max, learning rate 0.05, 63 leaves, min 100 samples per leaf, row and column subsampling 0.8, L2 = 1, early stopping 100 rounds (gate on average precision)
Threshold τ	grid 0.05–0.95, maximising post-processed F1 on the validation home
Thinning	keep ±60 min around washes and > 300 W extra load; 15 % of the rest, re-weighted
Feature selection	correlation > 0.97 removed; permutation importance (80k validation minutes, 2 repeats), keep > 1 % of max
CNN	121-min window, 12 epochs max, batch 256, Adam 10^{-3}, early stopping patience 3

AI tools used. Claude (Anthropic) was used to help implement and debug the notebook code. All code was run on Google Colab.

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Data: REFIT: Electrical Load Measurements (raw and cleaned), University of Strathclyde, CC BY 4.0 — https://pureportal.strath.ac.uk/en/datasets/refit-electrical-load-measurements/

Appendix A — Full feature list (65)

`agg`, `d1`, `d1_fwd`, `abs_d1`, `base_24h`, `above_base`, `above_base_rel`; for each window w in {5, 15, 31, 61, 121} minutes: `mean_w`, `std_w`, `max_w`, `min_w`, `range_w`, `z_w`, `above_min_w`, `steps_w`, `high_w`, `mid_w` (50 features); `high_prev_30`, `high_prev_60`, `high_next_30`, `high_next_60`; `hour_sin`, `hour_cos`, `dow`, `weekend`.